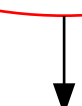
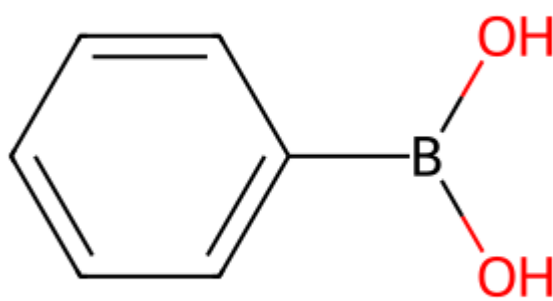


```
{'analysis_time': 0.0,
'first_solution_time': 0.306256,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 1.1615520884419839,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 1.1615520884419839,
'retro_star_value': 1.1615520884419839}
```

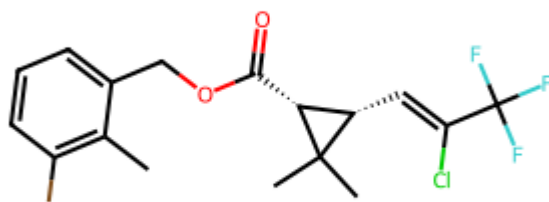


Reaction

```
{'analysis_time': 0.306256,
'first_solution_time': 0.672156,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 2.918104787874702,
'retro_star_reaction_number': 2.918104787874702,
'retro_star_rxn_cost': 1.8451602459551701,
'retro_star_value': 2.918104787874702}
```



```
{'analysis_time': 0.306256,
'first_solution_time': 0.306256,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 2.918104787874702}
```

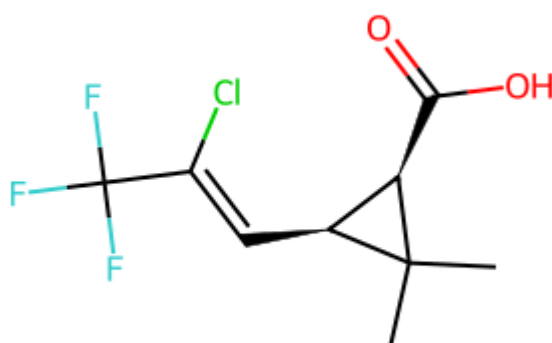


```
{'analysis_time': 0.306256,
'first_solution_time': 0.672156,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 1.0729445419195318,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 1.0729445419195318,
'retro_star_value': 2.918104787874702}
```

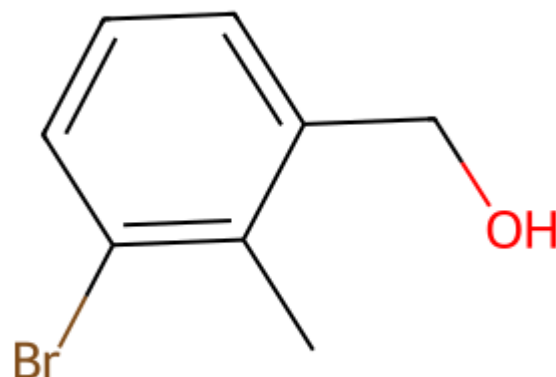


Reaction

```
{'analysis_time': 0.672156,
'first_solution_time': 0.672156,
'num_calls_rxn_model': 3,
'num_calls_value_function': 66,
'retro_star_min_cost': 1.0729445419195318,
'retro_star_reaction_number': 1.0729445419195318,
'retro_star_rxn_cost': 1.0729445419195318,
'retro_star_value': 2.918104787874702}
```



```
{'analysis_time': 0.672156,
'first_solution_time': 0.672156,
'num_calls_rxn_model': 3,
'num_calls_value_function': 66,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 2.918104787874702}
```



```
{'analysis_time': 0.672156,
'first_solution_time': 0.672156,
'num_calls_rxn_model': 3,
'num_calls_value_function': 66,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 2.918104787874702}
```